Portfolio Web Application

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About Me

Overview

A personal portfolio web application built to showcase projects, skills, and experience in a dynamic, interactive format.

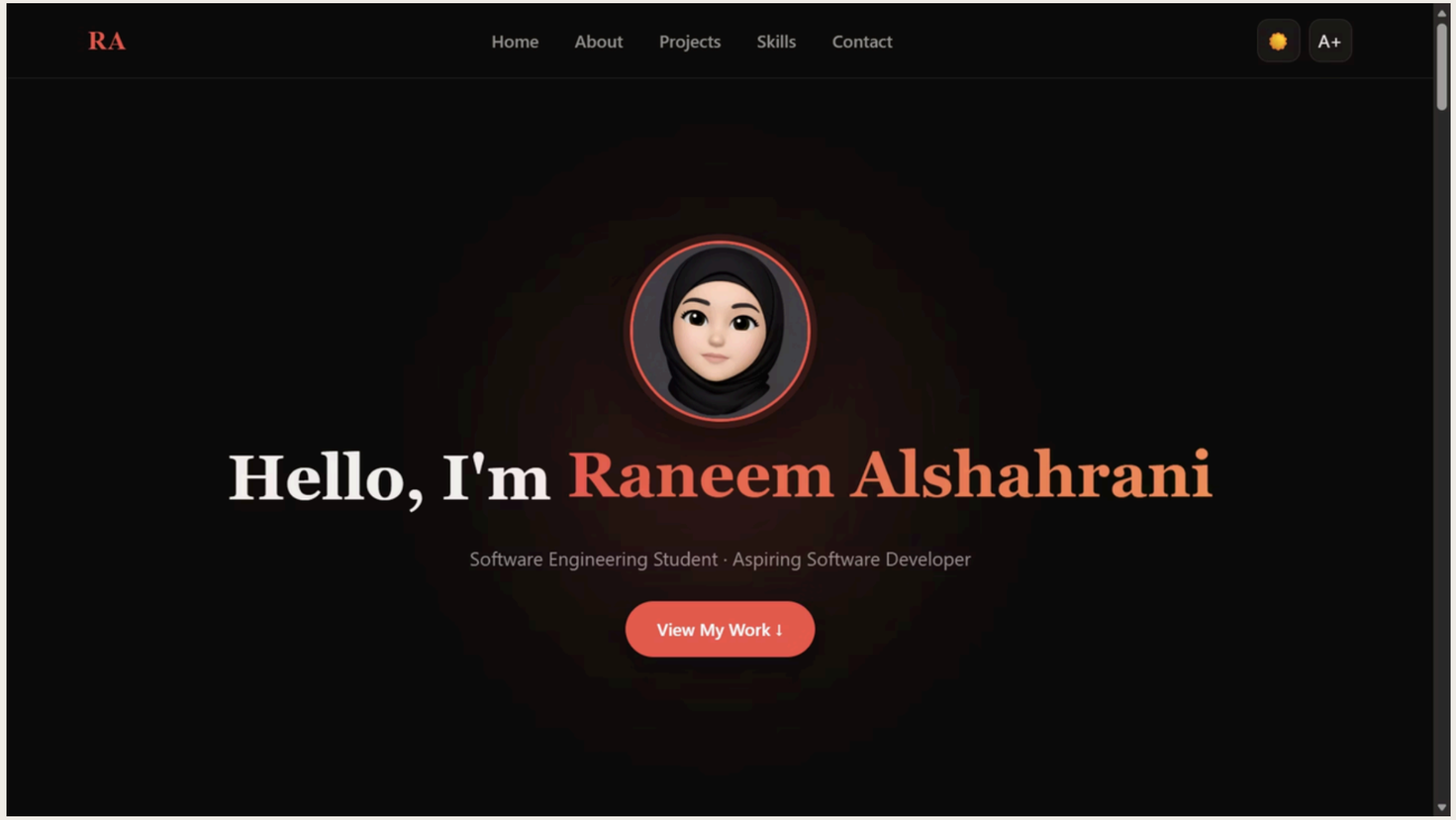
Objective

Demonstrate front-end development skills through a fully functional, real-world application — not just a static page.

Motivation

To go beyond coursework and build something tangible that reflects both technical ability and design thinking.

Technical Demonstration



Live Demonstration (Key Features)

- Responsive design (desktop & mobile)
- Light/Dark mode + font size control
- Project filtering and sorting
- GitHub API integration (live data)
- Contact form with validation

Technical Architecture

How It's Built

Single Page Application

All content loads on one page — no full reloads, smooth navigation between sections.

API Integration

GitHub API connected via fetch() for real-time repository data.

Web Storage

LocalStorage and SessionStorage used to persist user preferences across sessions.

AI Integration

AI Integration Highlights

- Used AI for debugging and problem solving
- Improved UI/UX design decisions
- Optimized code structure and performance

Tools Used

- ChatGPT (debugging, explanation)
- Claude AI (code review, optimization)

Core Technologies

- DOM manipulation & event handling
- localStorage (user preferences)
- sessionStorage (visitor timer)
- GitHub API & EmailJS

Technical Deep Dive

Challenges

- Achieving a consistent responsive layout across screen Responsive design issues on mobile
- GitHub API integration and dynamic rendering
- Managing filtering and sorting together

Solutions

- Resolved layout issues using CSS Flexbox and media
- Fixed layout issues using CSS
- Handled API data dynamically
- Improved UI interactions and animations

Unresolved

- No backend for storing messages
- Depends on external APIs (GitHub, EmailJS)

Planned Future Work

- Add backend support for storing messages and user data
- Improve UI design and mobile responsiveness
- Expand features and interactivity

Innovative Solutions

- Real-time GitHub data integration
- Visitor personalization (localStorage)
- Session timer and scroll progress bar

Lessons Learned

- Building responsive web applications
- Working with APIs and dynamic data
- Debugging real-world issue

Conclusion

Project Outcomes & Achievements

- Fully functional, deployable web application
- Responsive and interactive design across all devices
- Integrated real-time data via GitHub API
- Successfully deployed as a live application

Future Improvements

The next phase of development focuses on adding a **backend server** to handle form submissions and user data, as well as **enhanced UI features** such as dark mode, animations, and project filtering refinements.